

Reflective Report

1. Introduction

This is the reflective report for DES512, aimed at reviewing and reflecting on my work as an environmental artist during the creation of the game "Hell Of A Rescue" with Team SumoZ. Revealing how my work was efficiently converted to support better game performance, fast iteration, and contribute to the team's overall progression, despite the limitations of an unstable design direction, a restriction of not over 200 triangle counts per object, and the team's consistent fine-tuning of visual deliveries.

2. Technical Competency

2.1 From high-poly models to limits under 200 triangle counts

My original plan was to create detailed models with around 1000-2000 triangle counts per model, or even more. I have created many detailed models over the last 2 terms, which has brought me success in our games. This is based on my estimation that we are still able to make the game functional while maintaining the excess amount of detail. The plan is to strike afterwards, down to no more than 200 triangle counts, using low-poly models, which is best for optimisation.

Compared to making detailed models. The limitation of 200 triangle counts has given me another perspective to challenge: to capture the most iconic features of the object without making a super complicated art piece. It is a practice of observation and summarising. Benefiting from how I usually work, I can always return to the previous step when modelling, and several sets of 3d assets were remade from the 90s theme to the 80s theme this way. At first, I was trying to apply the multiRes method Euan told me about. It didn't work well and lost many details. I am thinking maybe it needs to be applied when the modelling work is at an early stage, and it only works on simple shapes with smooth surfaces to increase as many details as possible. Anyway, I realise the best way is to remake everything, and I started to work in the middle of week 3-4 probably. Took me 1 to 2 weeks to finish everything; I am proud of what I have made in such a short time, for they are easily recognisable from a further distance, while everything else is low-poly count. The whole process is more like a sculpting work rather than modelling. Especially when I was making the tabletop plant for the office.

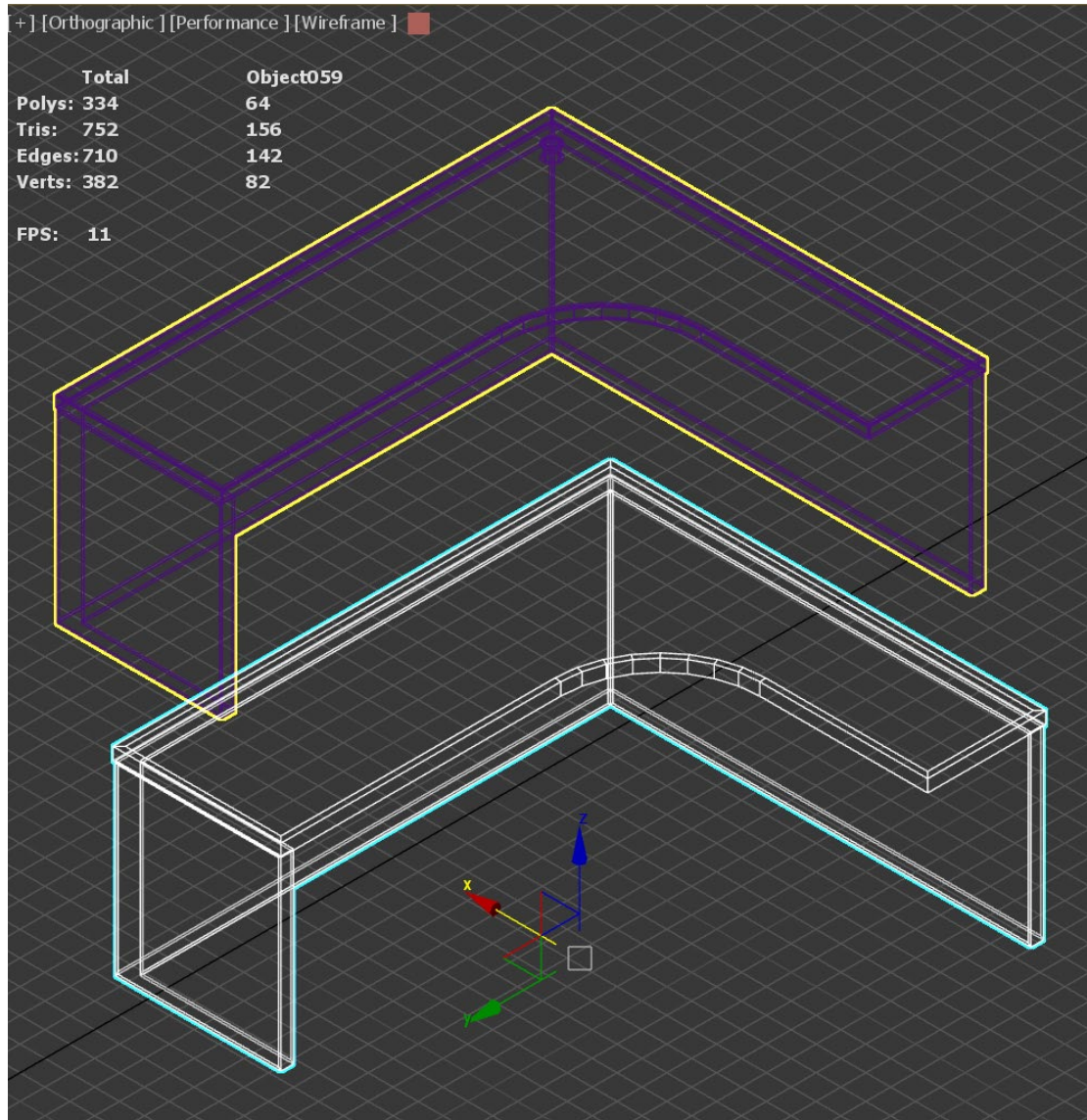
If you look closer, you will see there is no stem between each leaf. In fact, I took bamboo as the inspiration for the model and increased the number of leaves so that it looks more suitable as a tabletop plant. I made a group of triangles and resized them according to their relative position. The leaves at the back are bigger. I also left spaces in between them to pretend there is a stem. And then applied different modifiers like bend and twist. Before saving the changes

to the group of leaves, I always put them in comparison to make sure they don't always look alike. The randomised status of leaves can make the whole plant look alive. Overall, it looks like it is revived from a painting that captures the iconic part of how lively it is. It doesn't need a stem, and the shape already explain the functionality of it.



The next item is the office objects. Mainly the fixed position and tabletop objects. I did a similar thing last term, which was also an office model for the Cumbernauld town centre building game. I used the same model for quick ideation in the early stage, and eventually got far beyond that because the age doesn't match at all. I started to imagine the cubicle I'm about to create would have the same L-shaped layout just like Neo's office desk, removing pc from the table as the client doesn't want too many computer elements. Taken as an example from

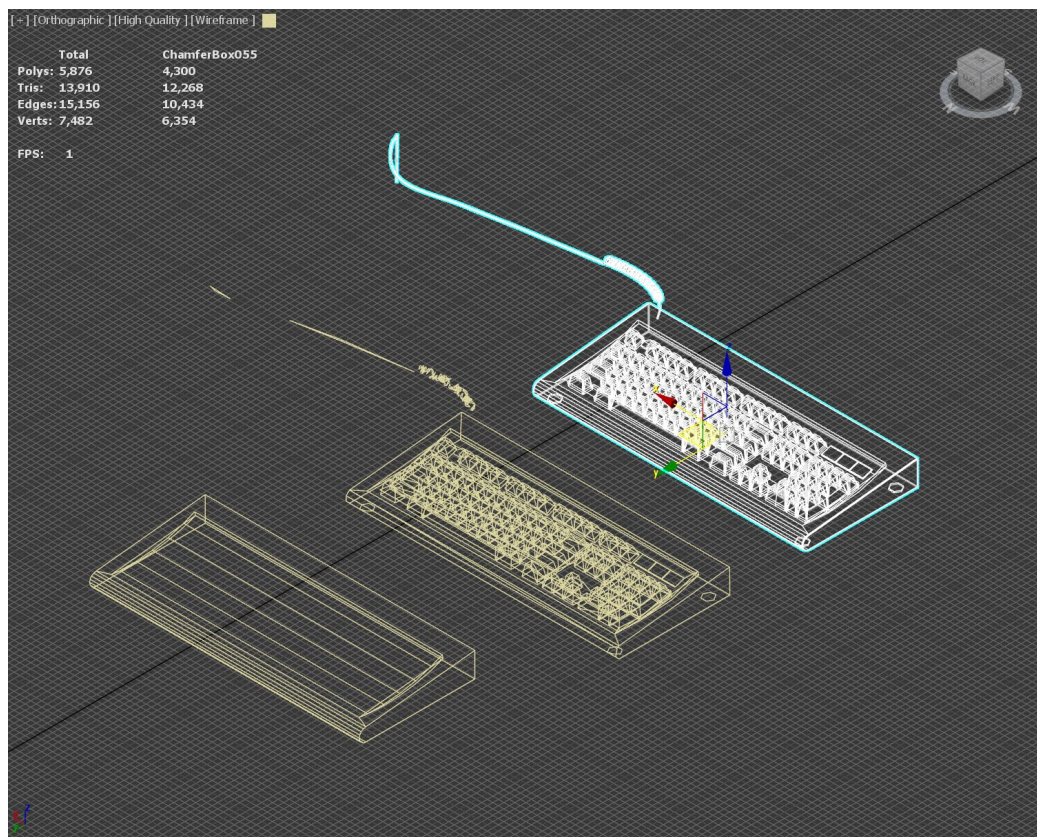
the L-shaped table, which exists in both the 80s and 90s versions. When I rapidly reduced the triangle count from about 600 to 156, it still kept its look in the camera settings in the orthographic view. But the space in file size it took, which only fit one 3d model once, is now fitting more models.



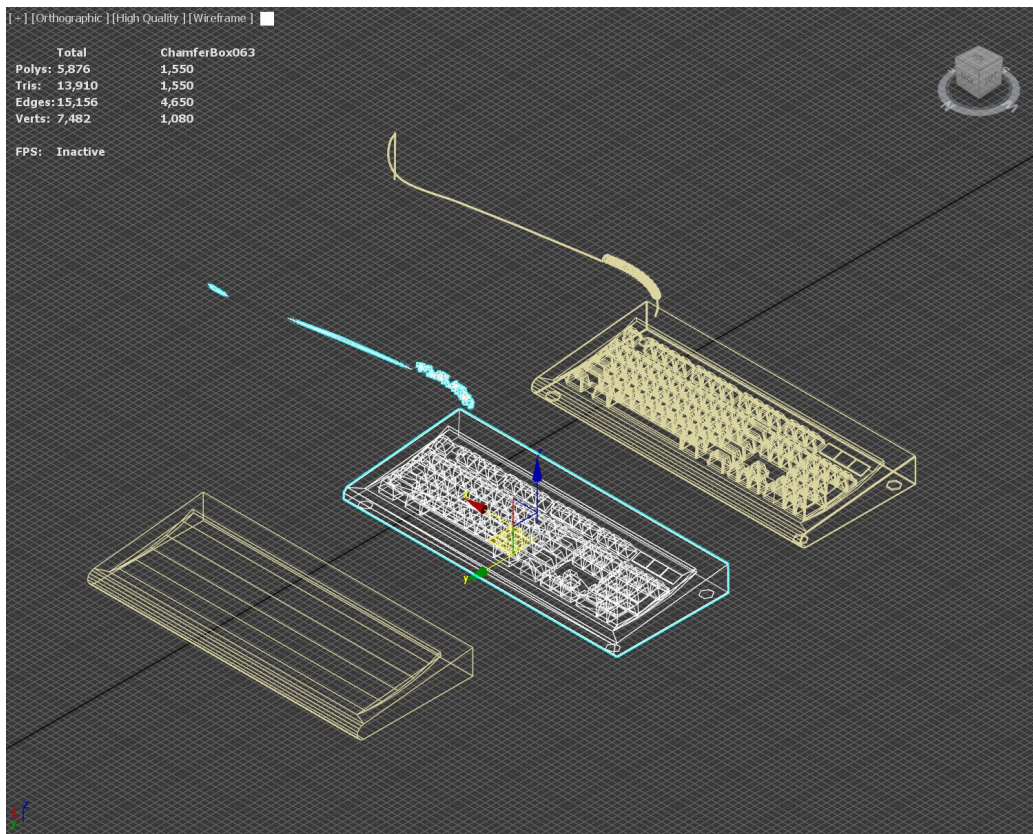
Comparing both tables, the upper one is the original and costs about 600 triangle counts. The bottom one costs only 156, which is significantly lower.

During this term, I have created many detailed models with significantly fewer triangle counts; although the 90s-themed objects were not put in use at all, the idea extends to the 80s-themed objects for how close they are in time. Sam mentioned there should be stationery on the table, so I got a stapler, a pen referenced from the BIC Krystal ballpoint pen, a tape dispenser, a file tray, file binders, a clipboard, and an ashtray suggested by Vera and David. For the heavy equipment, I have got pc set from the IBM famous personal computer, model M keyboard into low poly model and the mouse, 2 variations of a printer: one is tabletop, and

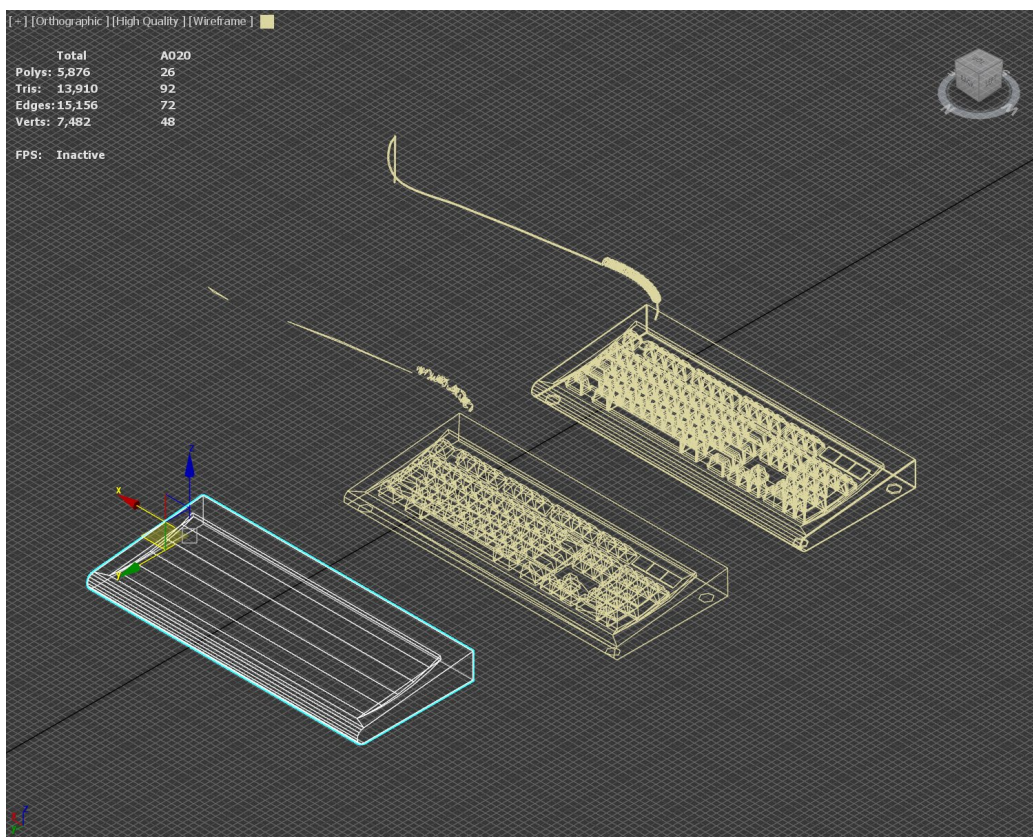
the other one is on the floor. For the Model M keyboard, it is where I started to try the high-to-low poly technique. The original keyboard model I created has excessive details and around 12000 triangle counts, for I recreated every single key with the curvature and the corresponding groove on the keyboard body. For the low-poly version, I failed to render for the first time in week 5-7 where I imported everything into the engine and Sam started to put them in the level. I was struggling with the use of the “projection” modifier in 3ds Max to perform render-to-texture. At that time, I didn’t know the modifier automatically handled the increment of the size of the cage by moving the contacting point along the normals of the corresponding vertices on the hard edges from the high-poly model. But I finally succeeded in week 10 or 11 and put that experience on the American Surety Building model as well, which is now perfectly cut up into different pieces and placed on each level.



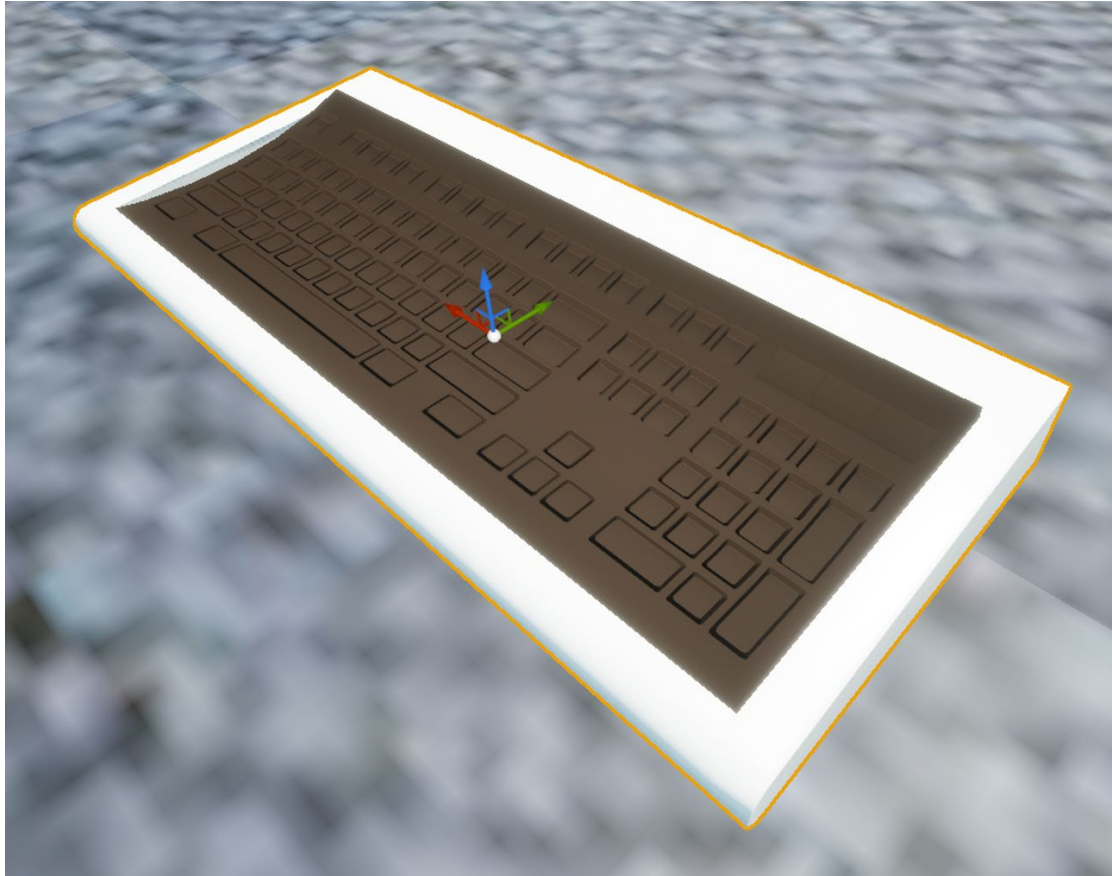
Original model M keyboard, costs over 12000 triangle counts



Improved model with multiRes modifier, triangle count: 1550



Optimised model for normal mapping, triangle count: 92



Final product

When I looked up the “Nutmeg!” reference after the client meeting. I quickly revised some of the outdated technology. The most iconic one, I think, is the visualiser. I don’t know if schools nowadays are still using the same thing for teaching purposes or if they are outdated for educational purposes as well, but they are popular in those old movies when it comes to the presentation part. The movie “Catch Me If You Can” recalls my memory. When the FBI agent played by Tom Hanks has to present the information he knows about the fraud played by Leonardo DiCaprio, he used a similar device, leaving the iconic circular light spot on the silver screen. Therefore, in the tutorial level, there is a visualiser on the desk as well, as the room is where HR Moira performs her introduction to any new employee.

The employees in this office would be living under pressure every day, and they can only drink coffee and smoke to soothe their nerves. They have a lot of work to complete so that the company can still maintain its large size. They have enormous stacks of paper on their desks, on the floor, everywhere, like some kind of haunted ghost. What’s even funnier is that Euan designed a vanquish animation for the employees, where they will end up vanishing in a shower of flying A4 paper.

It is the tearoom left to introduce. I started it off with the water cooler first; until now, I am not sure if I got the correct model from the 80s or not. There aren't many images on the internet to support what kind of water cooler was being used in the 1980s, and they have 2 main variations: one has a heavy glass jar installed on top, and the other one is what the game has. It has a swappable water jug design, and it is made of plastic. I think I also saw one image of a tearoom setting on Pinterest and thought it would be lively when the replacement water jugs are placed under the desk to save space. For the coffee maker part, I have actually got a whole vending machine for coffee as well. Look closer (if possible), and you can see a paper cup is placed at the dispenser of the vending machine. I was imagining the regular coffee maker on the desk provides cheap and nasty coffee (sorry to all coffee lovers, but I don't usually drink coffee), while you finally have a choice when you are willing to pay at the vending machine.

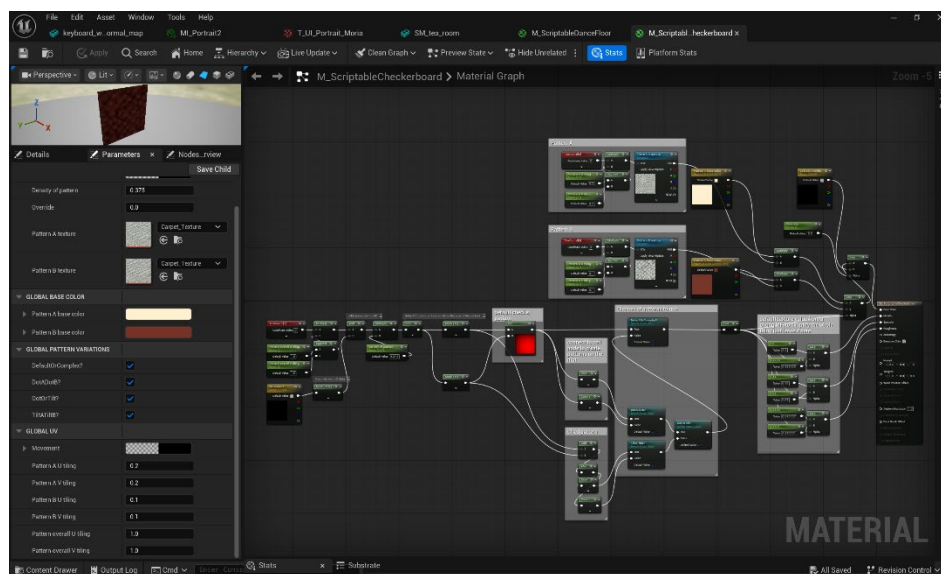


A small decision which brings not much difference in visuals, but now it allows our game to run so smoothly with fewer draw calls and less burden on the renderer, while fitting in more objects in the scene to make it immersive and engaging. So that we can support the theme of a "corporate hell" better. It is where I learned that art quality is not just about how many polygon counts I can get for the 3d models, but also cleanliness to see through the environment to finally get the message, balancing performance costs and the need for requirements.

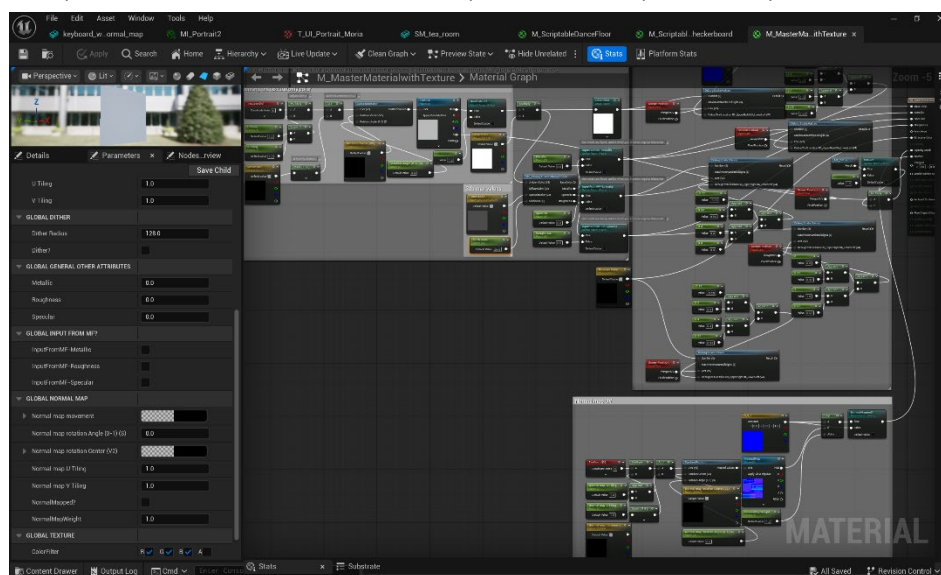


2.2 Material Instancing and scriptable checkerboard floor mat

For the texture and materials, I learned in Term 1 and have been using multiple material slots and combined everything well before importing into the engine. As I am more capable in using Unreal Engine, I have learned material instancing as well. Firstly, I created materials for every model with their corresponding metallic, specular, and roughness settings. Then I tried to optimise the render pipeline with the material instance concept. From what my term 1 project has got I have learned the basics and successfully replaced numerous materials with a master material and its numerous instances. Later, I will also talk about how the concept has extended even further, covering materials with texture, normal mapping with the dither effect using only 1 master material, and another application of this master material: the scriptable checkerboard floor mat texture with extended UV tiling and movement, setting all using material functions only.



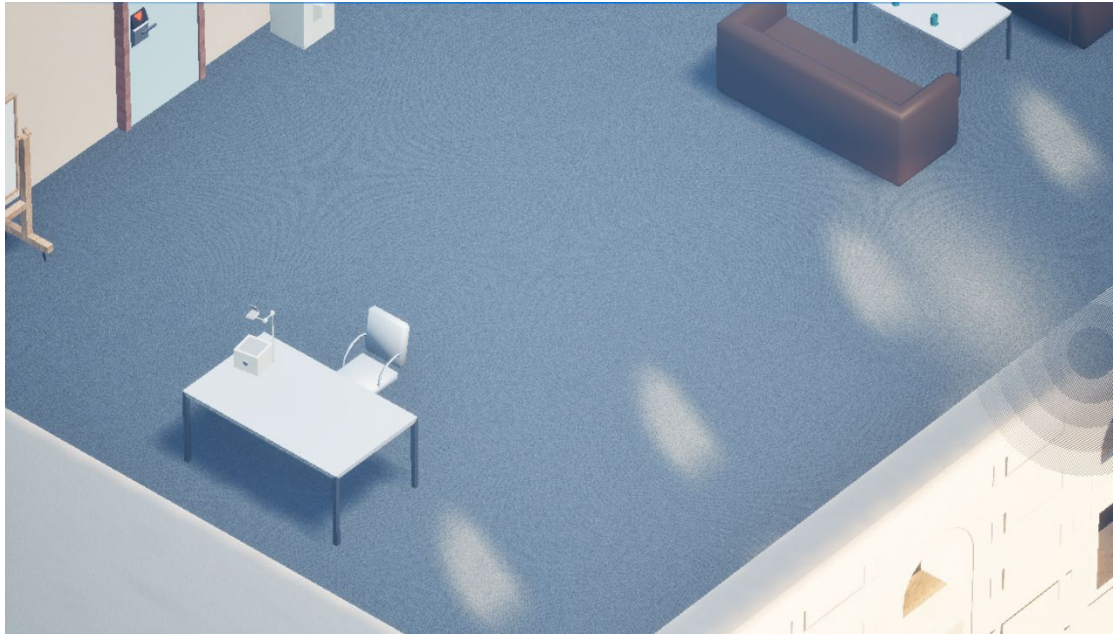
Scriptable checkerboard floor mat material, questionnaire-like parameter inputs on the left



Master material with texture, questionnaire-like parameter inputs on the left

After some play testing in the middle of development, I received some feedbacks mentioned the pattern of the floor mat is quite dizzy because of how frequent the pattern is. Sam was also requesting a final decision for the level 2 floor mat and also some additional objects like an elevator door. I took about 1 day to complete them and was thinking that if I had a floor mat texture controlled by a script in the game, it would be easier for anyone, including myself, to adjust it instantly to see any changes. I started working on that and initially created the prototype of the scriptable checkerboard floor mat texture. It has 2 material slots to put in 2 different textures in a checkerboard pattern (if anyone wants it could be the same texture or without a texture as well), UV related settings to control the tiling (later I even added UV movement that allows more control for the texture of the floor mat), and colour mixing function by masking the colour channels of the texture and manually added a new colour to the texture and eventually leads to the output. It is such a wise design that it has been used and evolved throughout the whole development process while being easy to understand. I have learned how to use a static switch parameter and how to use a group tag to put different options together to improve readability while editing the material instances. Now I can easily change the floor mat so that it is not so dizzy and apply another marble texture to make a difference from the regular office floor mat.

I was making the floor mat in 3ds Max before; it is not ideal for performance, but I can adjust any patterns I want. I have tried a few, but they could not be easily applied to the game. Using this method I will need to design or adjust the frequency of the patterns, measure for the size of the floor of each levels in Unreal, go back to 3ds max and slice the floor mat to match the size with precise x, y, z coordinates, assigned correct material slots and the multi-slots material to the checkboard so that Unreal doesn't automatically merge all unoccupied slots into one, move the floor mat to origin 0, 0, 0 and export. Eventually import into Unreal and rename, reassign material instances and clean up the unused materials when imported, and then I can use them in the level. This whole thing took 30 minutes or more, depending on how desperate I feel doing this. Moving on to scriptable checkerboard material in Unreal is another level; I only need to set up the material correctly once, and then Euan and I can already adjust the data to control how frequent the pattern is or any setting we need, taking only 10 seconds. No need to go back to external software to solve my problem, and we can compare the difference to be made instantly. I didn't receive any other feedback mentioning the same dizzy feeling again, proving that this material system I set up has solved the efficiency issues and improved player experiences effectively.



Before



After

The master material I have been using and the scriptable checkerboard floor mat material come with other improvements. Originally, I was using pure materials on almost everything, then I tried to move to using material instancing. I set up the master material in a way to adapt to the usage of textures and the ability to mix colour with the desired texture; I even adapted the unique calculations automatically generated when a file is imported from a 3ds Max file to Unreal. It is special because the way a 3ds Max object handles the material with Unreal Engine default import settings is via an external Phong function. Long story short, it turns out the metallic, specular, and roughness settings are completely different from assigning fixed

values. Without it, the effect of some materials will be greatly reduced. It is also where I implemented the UV tiling-related controls, including UV tiling and movement to enable fine manipulation of the repeating patterns. The same logic is also introduced to the scriptable checkerboard floor mat, with the experimental pattern generation using a different combination of sine, cosine, tangent and ceil functions in the blueprint. It forms the patterns to be used in level 3 stage 1.



Level 2 floor mat

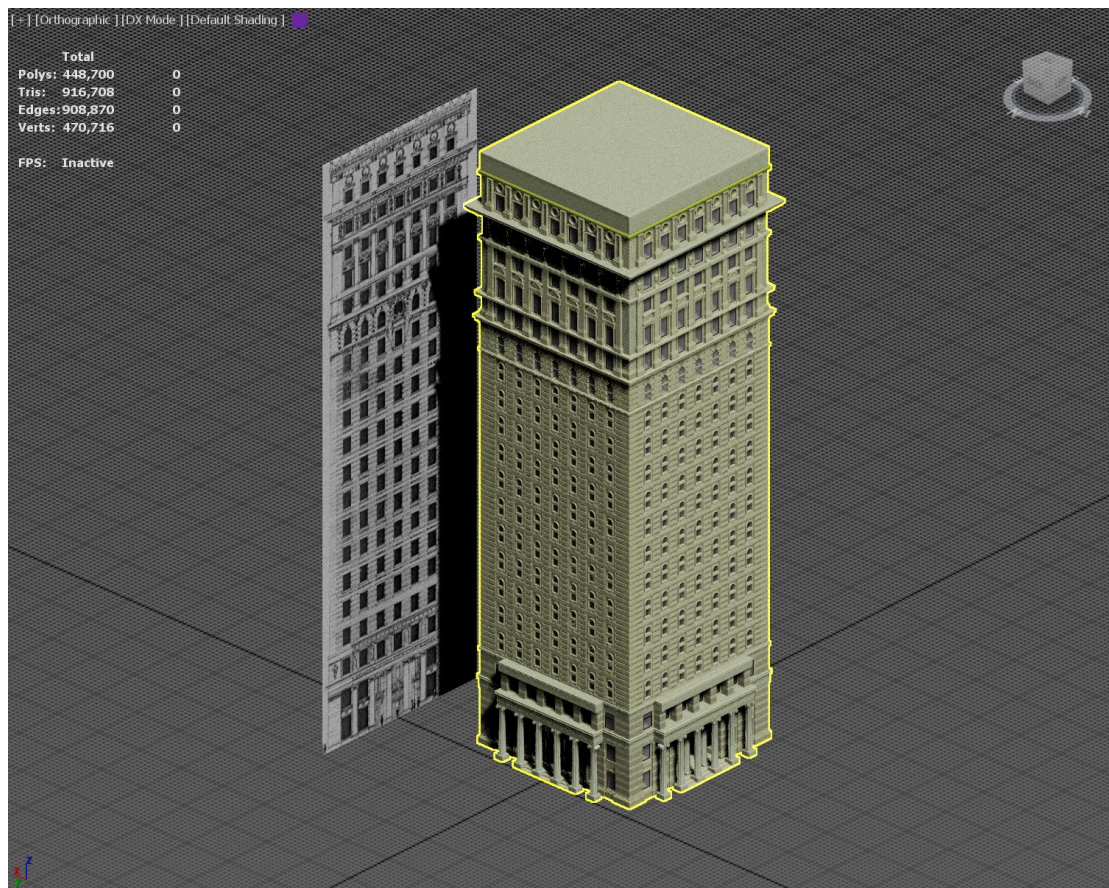


Level 3 stage 1 floor mat

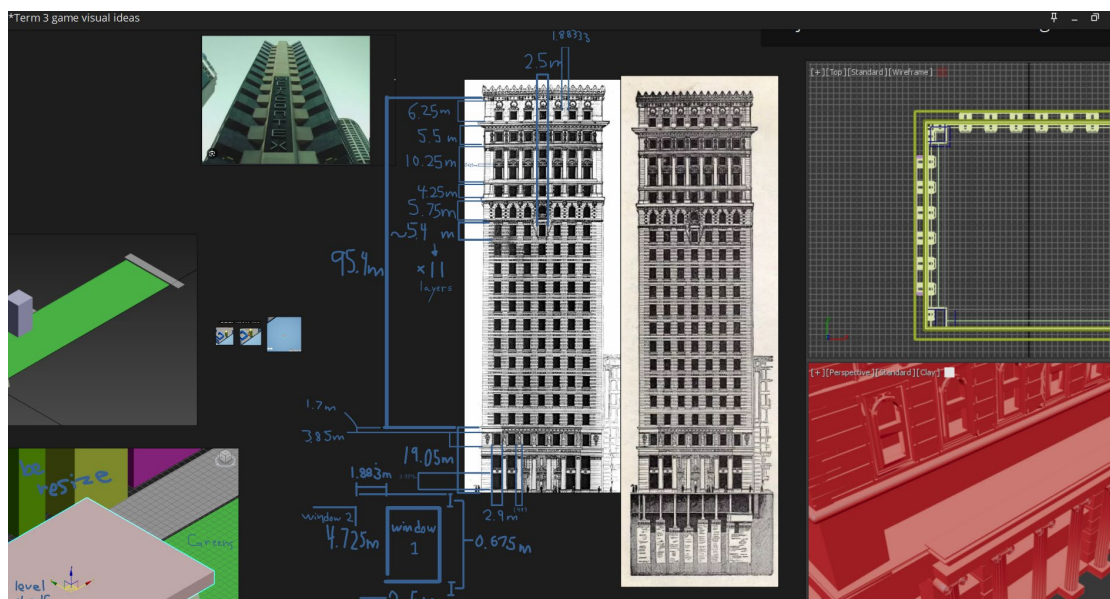
2.3 American Surety Building: Modular, UV unwrap and normal map

Most of the time I spent from week 7 to week 11 was on the creation of the American Surety Building. It is a building from the US and has existed for over a century, as far as I know. It is also a perfect target to use in our project because of how complicated and suitable it is for the incredible amount of ancient Greek decoration it has on the exterior; in fact, it is the only item that, when the client saw my alpha version of the high-poly model for the first time, he said he loved it. Not so many concepts we have gone through from different departments in all the client meetings had the same ratings.

The building was picked by Euan on one of the overworking days. We didn't go through too many targets to get to the building. However, the creation of the building is far more difficult than I thought. Through the reference research, I realised there are not enough details to recreate it, so I had to be creative, and I did. The building has been through many years, so of course it has been through reconstruction. Luckily for me, that doesn't affect the exterior too much, but only extends the building in its original exterior design style. It is divided into 23 floors, and I try to reference the original 7 windows x 7 windows version. The plan is to create details on the wall for one of the sides so I can duplicate them to the other 3 sides. The final high-poly product I've got has about a million triangle counts, meaning that I have aligned about 250000 triangles in 2-3 weeks.



Through observation, I realised the building has 5 main variations of exterior design that I can put to use in the game and create an illusion over 3 levels that they seem to be happening in the same building, and the level is getting higher and higher. Euan agreed with my vision. So I started working and began to measure the compartments first. As I measured the size of each compartment, I started to put a lot of boxes together as placeholders, as I also realised such complicated models have to be done by creating corresponding detail models first and combining them eventually. My reference image is blurred, and there is no way to find a clear image for me; there are some clear sketches on the internet that only contain part of the details, though. Most of the time, I am using approximate data from visuals and different references.



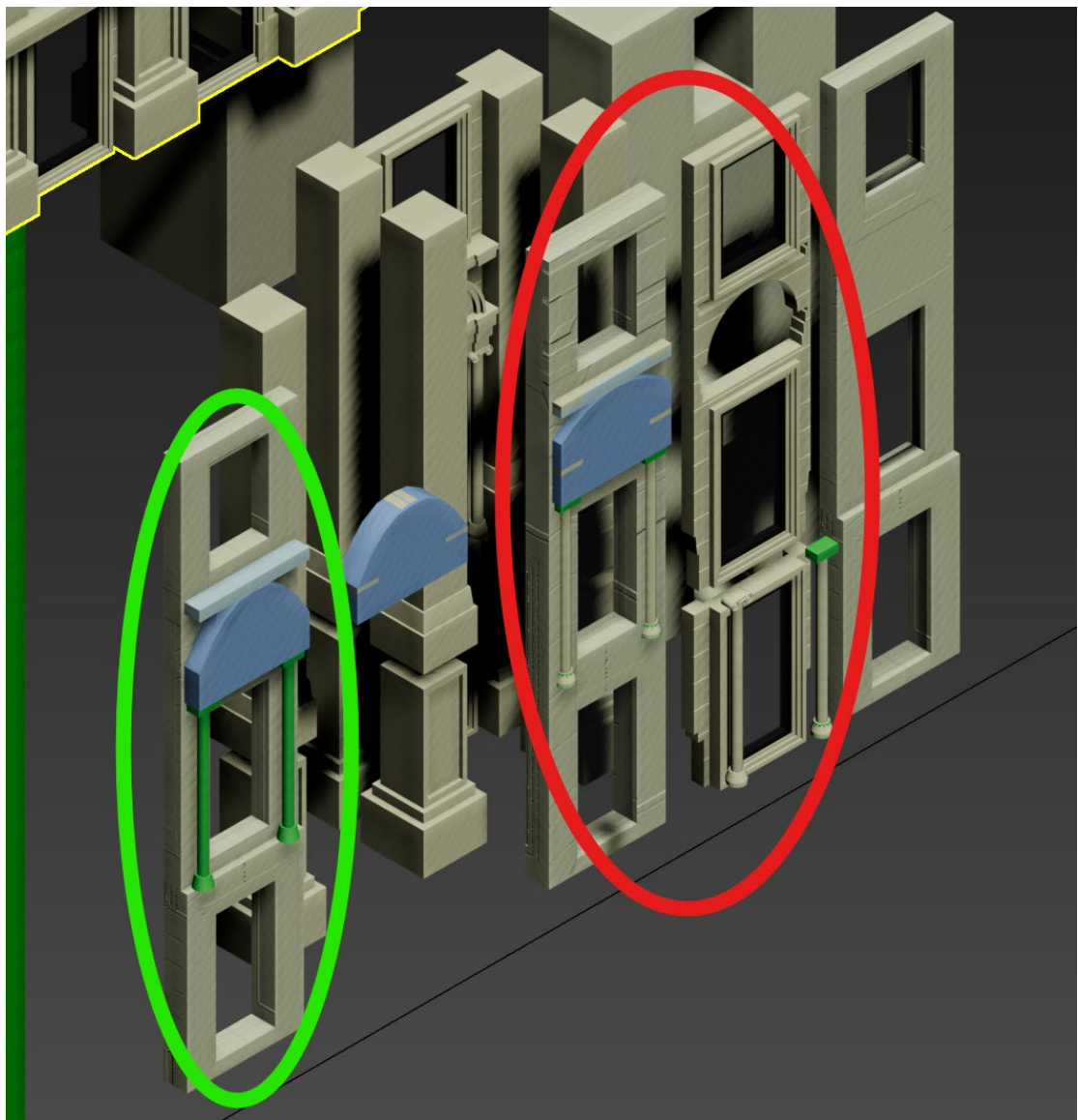
Measurement of the dimensions of the final product

I separate the whole process into different stages, as my way of recreating the details will make my working view very messy, which is necessary for me. I started with the middle 11 floors in the first stage. The same pattern is replicating, completed it will allow me to have a better vision of the whole building. I separated the design into the layout “pillar<->window<->wall<->window<->wall<->window<->wall<->window<->wall<->window<->wall<->window<->pillar”. 2 pillars with 7 windows in between, and numerous walls to fill the gap. The same structure is extended to the other floor. The render-to-texture procedures went well; I took a trial on a raw low-poly model, which is a box sized the same as the cutout of the wall. The projection modifier is caging well, a bit bigger than the high-poly model as intended, and the rendered normal map turns out to be fine. I tried to cut the shape of the window out on the low-poly model from the same spot as the high-poly model, and the cage

started to get tricky. Eventually, it also looked great the second time I tried to render the normal map of it.



It is also where I truly learned the real usage of the whole render-to-texture logic in 3ds Max. I believe the same logic is worth referring to in Blender. So things are getting complicated when I cannot easily just put a simple low-poly model and render everything to a normal map. On floors 16 to 18, there are several small but important external compartments like the pillars, the dome shape on top of the pillar and the extending window sill. I tried to use the same way I did last time, and the normal map is all wrong. I give in some time to investigate and eventually realised it is due to 2 main reasons. There is a gap between the pillar and the window (and so when the application tried to perform the render-to-texture move, the ray couldn't hit the back of the pillar and capture that part of the information in a wrong, distorted way), and the UV maps on the low-poly model are not unfolded correctly.



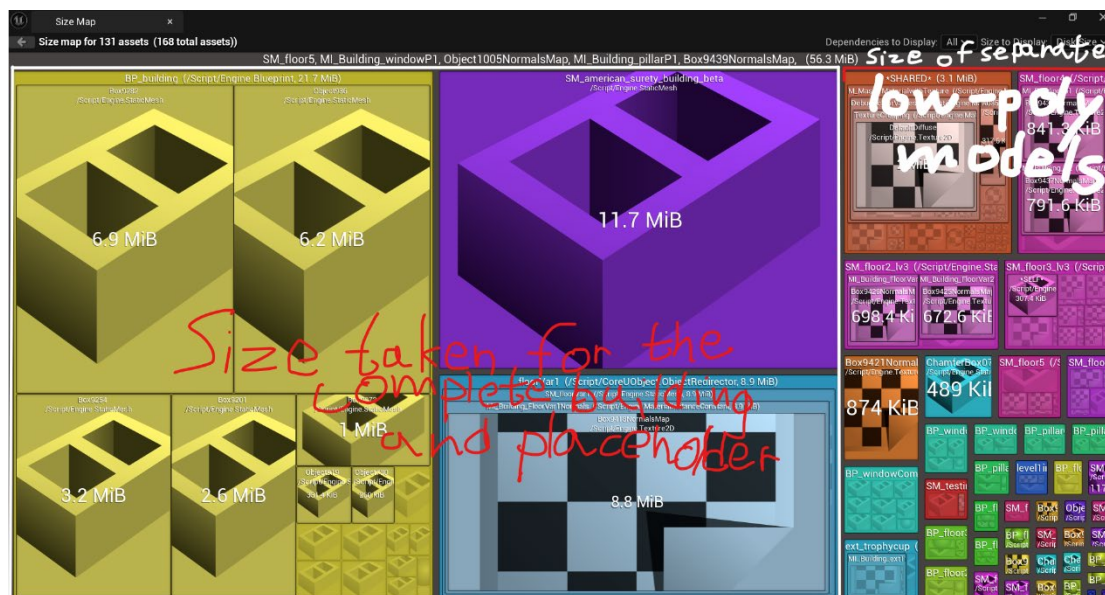
The detailed model (right one in the red circle) and the wrong low-poly model (left one in the red circle); in the green circle is the correct low-poly model which is thinner and gets the correct normal map.

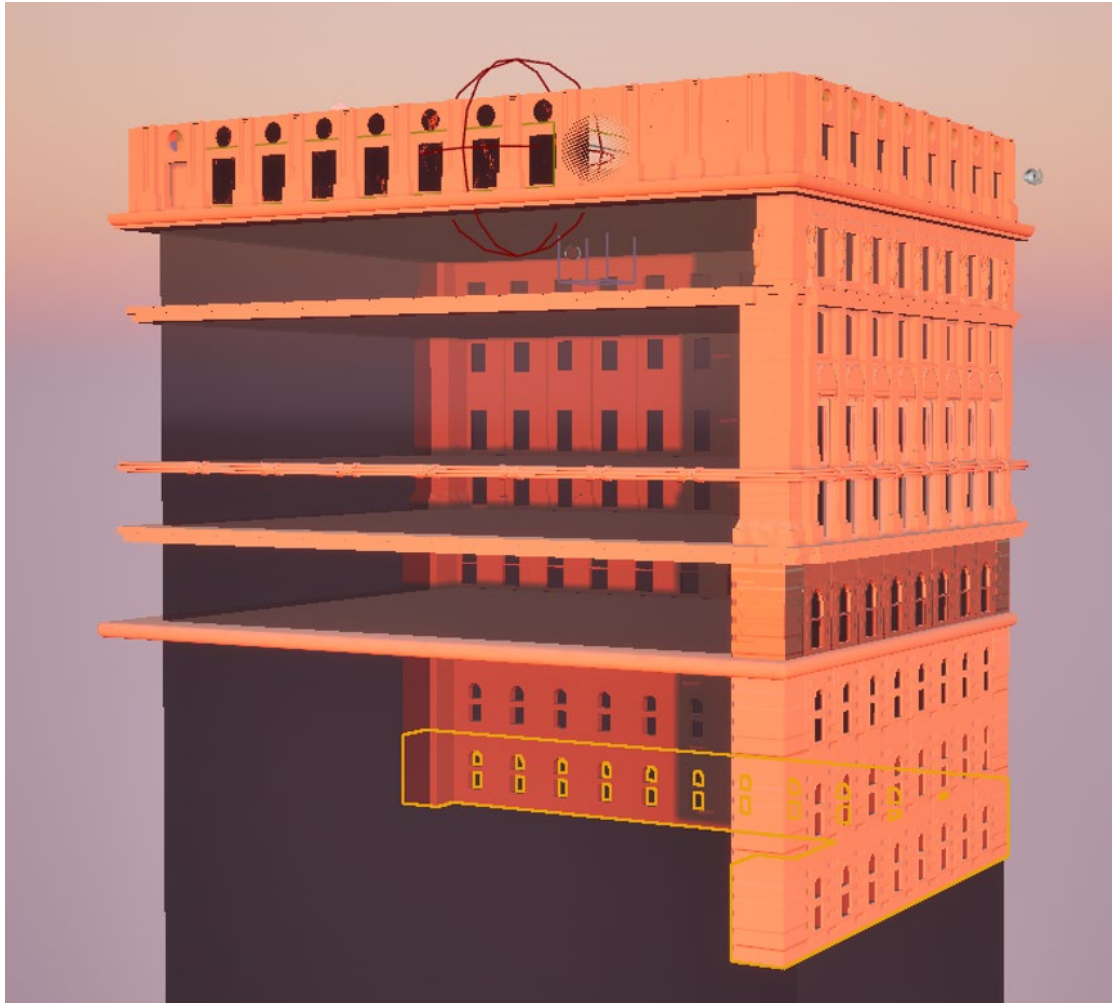


Final version in the game

The final move I take is to build up the corresponding low-poly model of each compartment along its hard edges. Practice makes perfect. I spent some time unfolding the UV wrapping of each compartment one by one. It is an art to perform with my sixth sense. I have to unfold them to get a clear mapping; sometimes they work best when unfolding along the angles, sometimes it is the smoothing group. I also have to apply a smaller resolution setting so that when all the normal maps in the game are added up for optimisation, they don't take up much space, but keeping enough details of the high-poly model to prove the artistic touch in these exterior decorations. Therefore, it is the art that could only be performed by the sixth sense. There was a time when I spent 50 minutes rendering an 8192x8192 normal map, just to make sure how much detail I needed in the game to balance with the details of my high-poly models, not even mentioning a bunch of wasted normal maps with different resolution settings.

Eventually, I get to keep each of the walls imported into the game with around 200 triangle counts and only 2000-4000 triangle counts when I put them in a group in Unreal so that they get the 90-degree view to use as exterior for a floor. Optimised normal map restricted to 1024x1024 for lower resolution lost its details significantly, while getting higher resolution can only increase its file size but provide limited visual improvements. I have kept the building at a relatively low file size compared to the original 91MB. I am also reusing all levels for the exterior, where the exterior of the former level was shown at the bottom of the current level. So that the player feel like they are moving upwards in the same building to maintain and improve the narrative between each level and visual consistency.





Behind the scenes, it is the back of the scene that the in-game camera will never see. It shows how the wall is constructed into a 90-degree complete exterior and is used and reused in level 3, stage 2. The same situation happened in other levels.

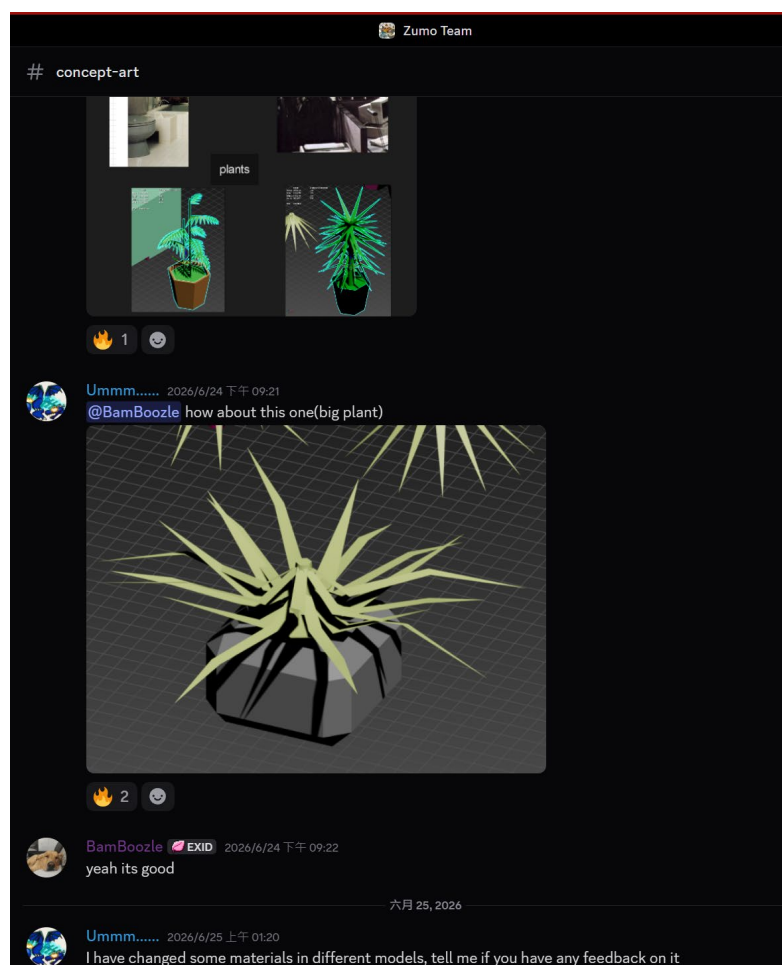
American Surety Building is the most complicated single asset in this project, in terms of visual perspective and time complexity. I felt honoured to be able to build it up as part of this game.



Followed by air con and an air vent as decorations on the wall. It also comes with a variation that has a half-cylinder-shaped display unit.

After the wall, I made a fixed version of the floor for the level to take place within; it is 50m x 50m. While added some stacks of paper placeholders as obstacles in the level. So the original look of the level is bigger and looks more like a jungle; the player will have to live their way out from enemies while doing the puzzle and area search. In the choice of colour scheme and texture, I used a blue- pale blue combination in use as the colour of the checkerboard floor mat. It also refers to the Matrix reference, but more because of the dull feeling; Euan loved the idea on the first day as well. Sadly, it doesn't seem to have been well delivered to the client. In addition, a complex Ionic pillar model and some different Greek patterns on the wall were created to help express the Greek theme.

Yet, the ancient Greek theme was cut from the game in one of the client meetings because of the confusion it could cause. Despite the chaotic situation in the first few sprints, our design team didn't make much progress with the client. I am still trying to perform fast prototyping and descriptive environment concept art, to try to help convert the abstract part of the art into a conceptual solution to discuss with the team.



As the ancient Greek theme is being abandoned, I shifted into the research of recreating a 90s-themed office, and it was working well. As I mentioned, I made numerous models at a photorealistic level. It is a challenge for me to see how much detail I can accomplish from some simple pictures from the internet. A necessary practice for any 3d artist who wants to make it a career. I was born in 2001, and I know nothing about the 1980s or even just the 1990s. They sound like BC 0 or the dinosaur age to me. Therefore, I must look up details on the internet to try to recreate them in the game. At first, I took The Matrix as my reference (this is why it was a 1990s-themed office) because it has a partially similar idea to our game. The big company where Thomas Anderson was working is a perfect corporate image for our game. High, tall headquarters, an image so bright that it can attract many people to die working inside, and an actual corporate hell image in the inner culture where everyone has lots of rules and regulations to follow (and therefore Neo is living in 2 lives). I kept numerous designs from the early investigation of the capacity for how well ancient Greek culture could be mixed with office elements. The ideas, like Ionic pillars, ancient Greek patterns, stacks of paper, window elements, as well as decorations on the walls, an elevator at the centre in the back, and the display unit, which turned into the trophy cabinets, were still in use.



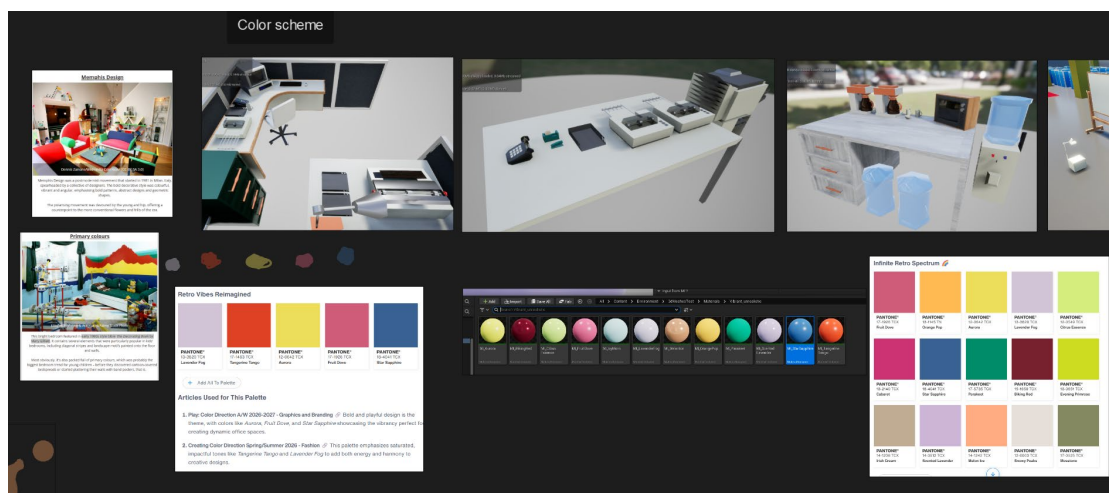
An office should be dull to highlight how the people inside are like walking zombies. However, the client doesn't like the idea, especially the involvement of too much computer in my design. I forgot what the reason was. He mentioned a game called Nutmeg! It has lots of 1980s elements for me as a reference, stating he wants something like an early 80s soccer manager's desk. I kind of know what that is, and it is about a small, dilapidated, and cramped office environment. It is not necessary to walk over to another cubicle, for you have everything you need in front of you, and kind of trapped at the desk. I was looking for a very detailed picture

of an office, and turned to research the small and cramped feeling of it. It changed not only the theme but also the priority of my central ideas: reusable, long-distance recognisable assets in our levels. These ideas are more important than keeping the photorealistic details.

This has inspired my design in general. I started to imagine the combination of the 3d assets in this way. Many 3d assets can form different cubicles in numerous variations and sizes. Not only that, but they can also form a little functional area by grouping related objects; for example, I put a TV next to a coffee maker in the tearoom set to form a functional tearoom where the employees will not need to walk around anymore. I imagine their attention could easily get trapped in such a simple preset of environmental objects.

3.2 Feedback and reusing assets

As my original target is to create a dull colour scheme office, I did make it dull, but the client wants the vibrant colour of the 1980s. On the recent mainland Chinese internet, there is a term I think best describes that vibrant theme. It's called “經濟上行時期的美” (how beautiful everything is during a period of economic upturn). Again, the 1980s is like the dinosaur age to me, and I don't get how vibrant it could be at all, so I had to ask AI again. This time I went to the Pantone official site, and surprisingly (or not so surprising, as everything is implemented with AI nowadays), they have an AI on their website. I then asked questions, and the general feeling was that it could explain to me and try to generate different colour palettes for that. One of the producers from another team, Jonpul(sorry if I misspell his name; I mean the charm-looking, always-smiling dude), came over and see what I had done and told me it is a brilliant move to ask advice from AI for the design palette like that. Then I put the hex colour codes in Unreal and left a few vibrant materials for Sam to use in the folders, if anyone seeks a change of colour in the 3d assets.



In the final development sprint 6, I worked with Euan on the final visual deliveries in every level. After Euan finished most of the concept art and 2d drawing work, I had him make the final decision on the colour scheme. This is where my scriptable checkerboard floor mat material was put to use. We reverted the level 1 floor mat back to the original blue checkerboard scheme, redesigned the floor mat colour and pattern for levels 2 and 3 to match the overall theme and characters in both levels. A famous quote said, “A diamond is a chunk of coal that did well under pressure”. I feel like this is the sprint where we both get the most creativity. I aligned the exterior well of each level. Looking at the pillars (they were all circular or rectangular, but now I can use render to texture to put a relatively low model of my beautiful Ionic pillar, which obviously could get better visual deliveries) and replaced them with the low-poly Ionic pillar. And then the Greek pattern I thought I was never going to be able to put to use for the Greek theme had been banned back in sprint 1. Eventually, I rendered a low-poly version of it to put to use in level 3, stages 1 and 2. All these elements form the immersive feeling of each level and turn it into the final product that matches everyone’s expectations.

In the 80s office placeholder assets, I’ve grouped different 3d models into 6 different cubicles. The best cubicles own 2 desks in a 3x3 space: one with a pc on top, and the other smaller desk has a typewriter; it even has its own printer. The worst one is just taking 2x2 spaces, using the smaller desk from the best cubicle, and maybe doesn’t even have its own typewriter (which is actually already common in 1980s offices). They can be seen across the levels. Sam has pushed this even further by setting up the 4 enormous working sections in level 3 stage 1. Very good job.



Reusing assets across all levels

I have successfully reused the same set of environmental assets in different levels to maintain consistency in the visual design language. Reduced the need to remake a new batch of assets when the requirements were always changing. Whenever an edit was made, material instancing always brought us the convenience of fast-paced visual testing and immediately brought us to the final decision when visual integration in levels.

4. Behavioural Competency

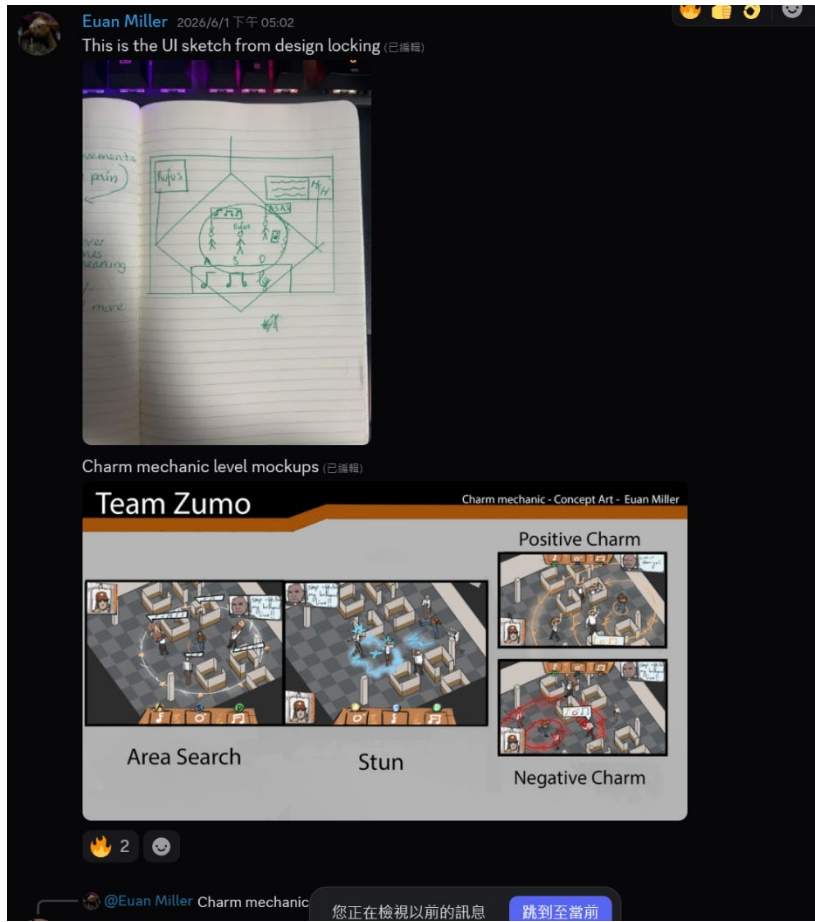
4.1 Collaboration under restrictions

Hell of a rescue, as a 2.5D game, 3d part is not really an important element in the game. It is more about how we turn our assets into something best optimised for better performance review, fewer draw calls, and a higher frame rate. Euan did a great batch of artwork in character animation by turning each character animation in the game into 16-directional sprite sheets on the first day. He saved a lot of time by rendering these animations in Blender and then import to the game. He has been doing it throughout the whole course, turning his art into sprites using a flipbook feature in Unreal. My way of working out the 3d assets is totally different in comparison. In the first thinking, I was aiming to create photo-realistic, or even digital twin level of 3d models in the game. It is the way I worked with the 90s-themed office objects and early-stage ancient Greek element objects. Technically, it works, and my desire is fulfilled, but in reality, such complex models are too much for our game and too long to make.

We never discussed how detailed our camera in the game will focus on the level, nor the scale of the objects. Upon completion of the environment development, the walls in all levels and the base underneath are still different from each other. It became a challenge in the early stage until Euan and Sam decided and told me that they wanted to keep everything simple by setting a limit for each 3d objects to be at most 200 triangle counts, or so. That means our game will be cartoonish in some way if we want to allow the player to look into the details. I am happy that I don't have to get involved in the team head discussion involving the standards of each visual element, actually. And their discussion swept away the obstacles for me. In the meantime, Richeng had also provided a lot of ideas that still affect our game until now. The character design and the contrast in the self-image of each character formed our initial impression of the game's graphics. I had a happy collaboration with them.

Ideation is a result, the summation of observation. During the creation, Euan introduced a few references from different TV shows, movies, etc. He didn't have time to create concept art as we did in Term 1. I am on demand of my own. We work in the reverse way; usually, it is for me to decide how the items look alike, and then Sam will put all the placeholders together. Euan will draw the concept art using the placeholder map layout and how my 3d assets will

look eventually. When I am writing this paragraph, I looked back to compare the final product and the concept art of level 1 by Euan from 1/6/2026. They look surprisingly similar, considering how many times we had to change our design, from the visuals to our gameplay design. It is encouraging to me that I still got to keep many items in my original design purpose, and many people agree with that.

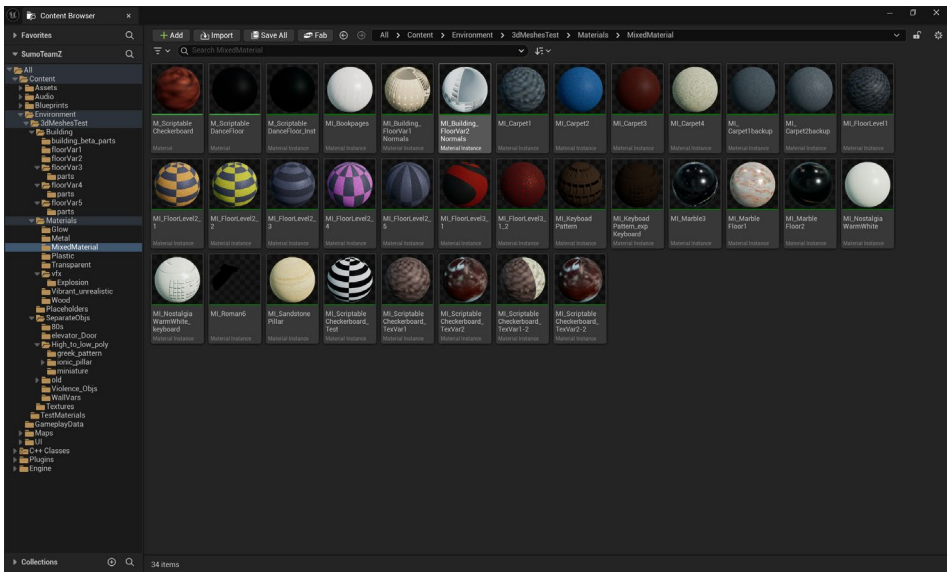
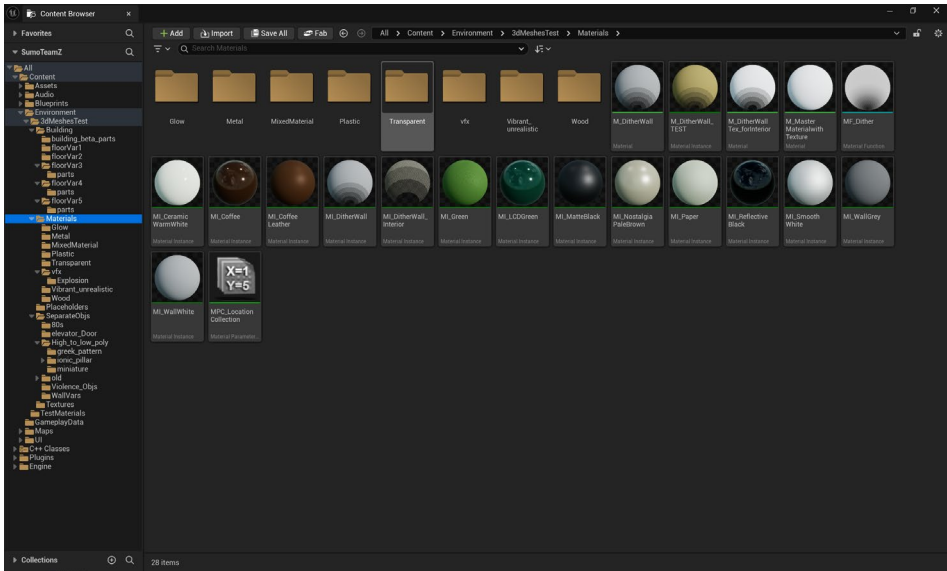
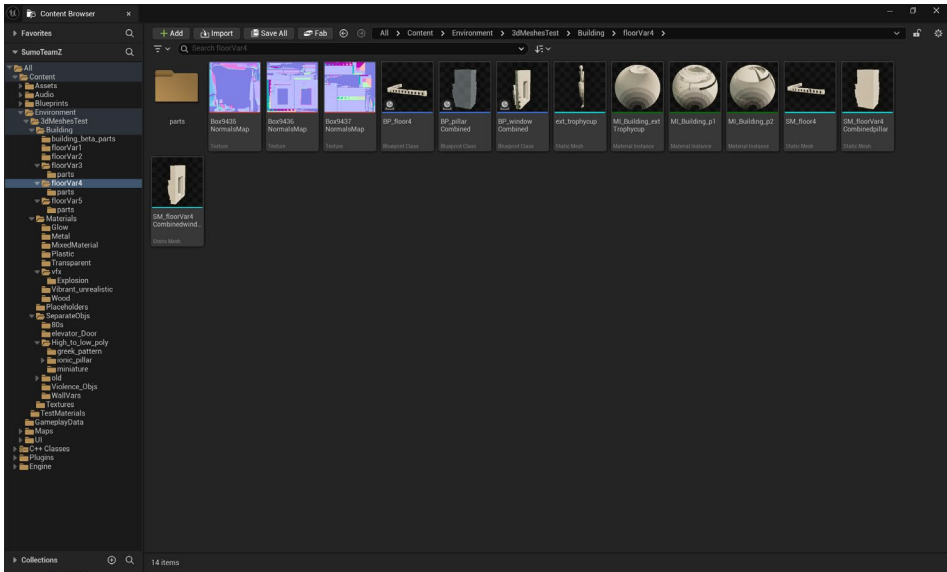


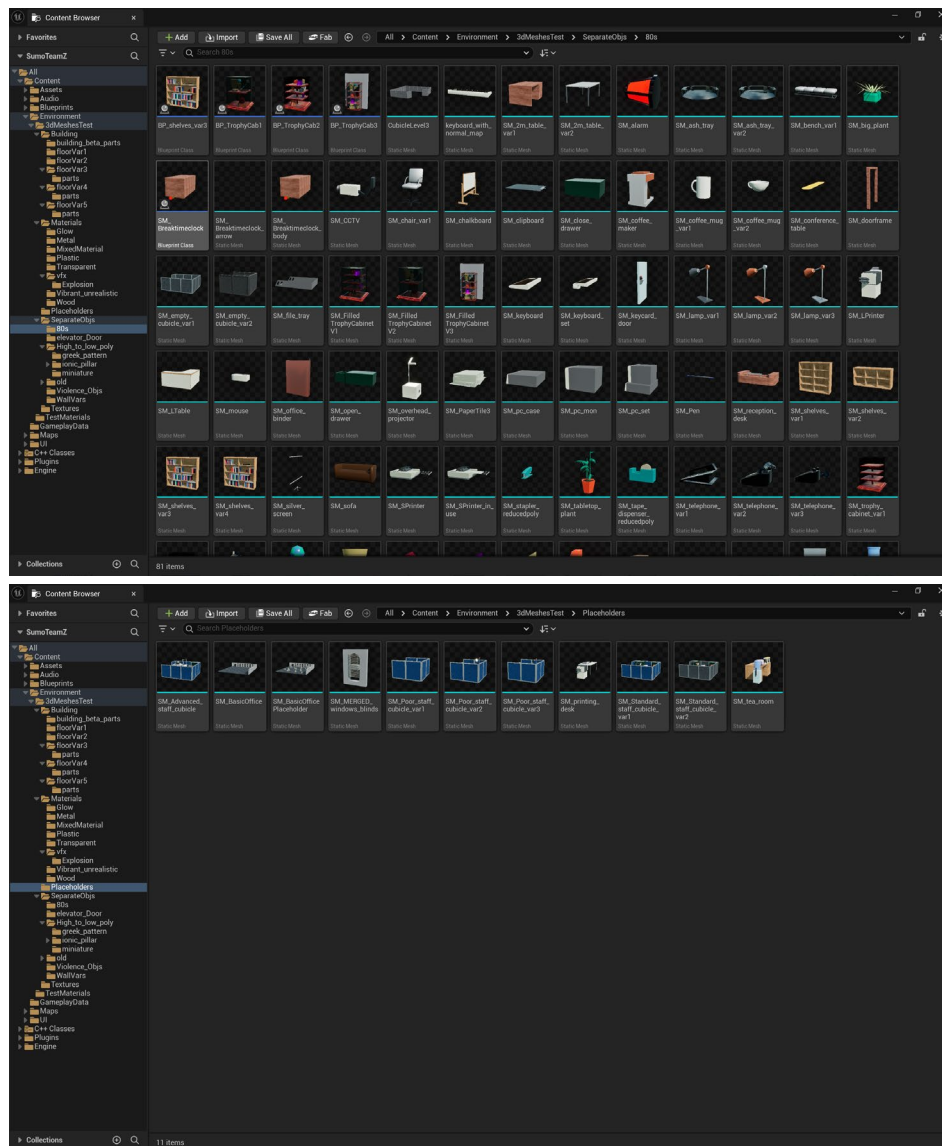
Screenshot of the early-stage concept art from Euan, after I have finalised the final look of the office environment



Level 1 overview, after week 12

I am not the final decision-maker in the visual design solutions of the overall game deliveries. My main role is to ensure the confirmed constraints in visuals from clients and the nature of our game can be turned into executable results in environmental art effectively. When the team decided to limit 3d assets to around 200 triangle counts each or sought better results in the material solutions, I adjusted my way to adapt and sought room for further development in order to achieve better cleanliness to see through the environment and improve asset utilisation efficiency. As mentioned above, everyone can easily access and use my assets in many places as well. The team, most of the time it is Euan and Sam, can quickly adapt to the questionnaire-like interface of the material instances and the easy-accessible 3d assets from a well-organised folder structure with proper naming.





The use of a clean, organised folder structure for collaboration across different teams for easy understanding

4.2 Personal skills assessment and future development

I am satisfied with the results I am obtaining from my work this term overall. Especially when it comes to the part where I convert the 3d assets and materials into reusable assets, how I design the scriptable checkerboard floor mat material for my team to access and recreate for iterative design, and the outcomes of the 3D models of the American Surety Building. By making all this work possible, I've learned to make low-poly models with a relatively light model to test the UV and caging, to advance the whole modelling pipeline. In the future, I will use the same and better methods in iterative design. Replacing complicated but not needed assets with optimised models for performance balance, and testing before beginning to work on those great, massive parts of the work.

5. Conclusion

This project demonstrated my ability to adapt environmental art production to technical, procedural and collaborative constraints. Technically, I reduced asset complexity to meet the 200-triangle limit, developed reusable material-instance workflows, and applied modular modelling and normal-map baking to retain architectural detail within an efficient budget. Procedurally, I responded to changing client direction through research, rapid environmental prototyping and asset reuse. Behaviourally, I supported the team by producing accessible assets and material systems that Euan and Sam could adjust during level integration. In future projects, I will test low-poly models, UVs and baking cages earlier in the production process before committing to complex assets, preserving the iterative approach that was effective in this project while reducing late-stage technical risk.